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Cyclic hydraulic fracturing of pocheon granite cores and its impact on breakdown pressure, acoustic emission amplitudes and injectivity

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ABSTRACT

Cyclic hydraulic fracturing, which employs cyclic injection with alternating high and low injection rates or pressurization, is suggested to assist in reducing induced seismicity. Our laboratory work demonstrates the impact of cyclic hydraulic fracturing on tensile-dominated fracturing of intact Pocheon granite core specimens. Cyclic injection of water reduces the breakdown pressure by up to ~20% and the maximum acoustic emission amplitude is reduced by ~14 dB on average. We observe hydraulic fractures in granite specimens through using computed tomography. Cyclic hydraulic fracturing creates complex fractures with more branches and smaller apertures compared with those created by continuous injection. The average injectivity of the fractured specimens by cyclic injection is smaller than the injectivity measured on specimens fractured by continuous injection.

1. Introduction

Deep geothermal energy is a renewable energy source with great potential for future use. However, intact rocks deep underground are usually of low permeability; therefore, hydraulic stimulation is necessary to create a permeable reservoir for extracting geothermal energy through water circulation. The reservoir is defined as an Enhanced Geothermal System (EGS) reservoir. Two hydraulic stimulation mechanisms can be distinguished that lead to EGS reservoirs: hydrofracturing (i.e. creating new tensile fractures) and hydroshearing (i.e. inducing slip along pre-existing fractures and faults).[1] Such stimulation can induce seismic events, which is a major concern and source of public disapproval.[2–5] For example, the EGS project in Pohang, South Korea, was stopped due to a M_w 5.4 earthquake in November 2017, whose epicenter was close to the drilling site.[6,7] Field data collected from EGS sites have shown that stronger seismic events are more likely to be induced by hydroshearing, i.e. slip on an existing fault.[2,3,8] Fault activation triggered by hydraulic fracturing was also reported during EGS and shale gas development.[7,9] Our primary challenge is to create a fracture network which achieves the best hydraulic performance of EGS reservoirs, under the premise that hydraulic stimulation treatments are safe.

New hydraulic fracturing (HF) techniques with minimal environmental impact (i.e., controlled and mitigated induced seismicity) are of great interest for oil and gas engineering as well as for EGS. New concepts have recently been presented, such as cyclic stimulation[10–13] and multistage stimulation.[14,15] In contrast to conventional HF, which employs continuous injection at a relatively high injection rate or pressure, cyclic hydraulic fracturing (CHF) employs a cyclic injection scheme comprising cycles of alternating high- and low-injection rate or pressure injections. Accumulated injection pressure is therefore frequently lowered to allow stress relaxation at the fracture tips. Zang et al.[10] noted that the failure process during cyclic mechanical loading can differ from fatigue hydraulic fracturing, where failure is driven by high-pressure fluid in a fracture, particularly at the fracture tip. Mechanical fatigue behavior is highly dependent on loading frequency as well as the material type, including its microstructures.[16–18] The evolution of existing microcracks in granites is a direct cause of mechanical fatigue failure.[19,20] The main difference between fatigue hydraulic fracturing and mechanical fatigue failure in granites is the fluid–rock interaction within the microcracks during fracturing and geochemical processes.

In recent years, CHF has been demonstrated at various scales employing different injection schemes, allowing the quantitative

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evaluation of permeability enhancement and the reduction of induced seismicity.[10–13,21–36] In a laboratory, we conducted a concept study on Pocheon granite containing preexisting microcracks.[21–27] For the first step, as described in detail in this study, we tested intact Pocheon granite cores under triaxial stress conditions to investigate the impact of CHF on the breakdown pressure, acoustic emission amplitude during fracturing, and hydraulic performance of fractured samples. Second, we tested Pocheon granite cubes under true-triaxial stresses, comparing different injection schemes including CHF and pulse pressurization.[23–25,27] Both tensile and shear fracturing were observed based on acoustic emission analysis.[25,27] The shear fracturing is probably due to slip on preexisting microcracks in the granite or on the newly created tensile fractures. For the next step, we plan to conduct an injection-induced shear slip test on the same Pocheon granite core containing a fracture plane to investigate impact of cyclic injection during the hydroshearing process. Furthermore, we conducted mine-scale experiments at the Äspö Hard Rock Laboratory (HRL),[28,29] which were accompanied by laboratory experiments on three different granitic sample cores.[30] At field scale, demonstrations of CHF have been reported at the Pohang enhanced geothermal site.[12,13] Zang et al.[11] and Hofmann et al.[31] reported a comprehensive comparison and further analysis of the above findings. Moreover, we performed numerical modeling at the field scale in order to understand the underlying mechanism of the different fracture patterns and induced seismicity between continuous injection and cyclic injection.[10,32–36] Numerical simulation using a 2-D naturally fractured reservoir model (2 km × 2 km) showed that the total number of induced events as well as the occurrence of larger magnitude events decreased.[10]

In general, cyclic injection can reduce the maximum magnitude of induced seismicity compared with continuous injection, although it seems to be less effective in enhancing permeability. In contrast to the aforementioned findings for granites, Patel et al. (2017) reported that CHF increases the fracture permeability of Tennessee sandstone by a factor of three to ten compared with conventional HF.[37] Their results, based on the location of acoustic emission (AE) events, also reveal that wider damage near the injection point is created by CHF. However, they did not report amplitudes of the events in their study. Similarly, compared to HF, CHF has been reported to reduce the fracture initiation pressure[38] and to create more fractures[39] in concrete and cement materials.

In this study, we present the first laboratory demonstration of CHF causing reductions in breakdown pressure (BP) and the maximum level of hydraulic fracturing-induced seismicity in intact granite cores containing pre-existing microcracks. In addition, we observed hydraulic fractures using X-ray computed tomography (CT), and evaluated permeability enhancement based on the apertures of induced fractures and the injectivity estimated from injection tests.

2. Concept of cyclic hydraulic fracturing

Hydraulic stimulation in the field is usually controlled by the injection rate or volume. Conventional HF stimulates reservoirs through continuous fluid injection to achieve a target injection pressure that ensures rock breakdown. Thus, the pressure increases monotonically. We define the breakdown resulting from continuous, or monotonic, injection as monotonic breakdown, and the corresponding injection pressure is therefore the monotonic breakdown pressure (MBP). In contrast, CHF employs cyclic injection, with alternating high and low (or zero) injection rates or injection pressures, in which case the resulting cyclic breakdown occurs at the cyclic breakdown pressure (CBP). Both continuous and cyclic HF can employ various injection schemes.[11–13]

In a laboratory hydraulic fracturing test on rock samples, fluid injection can be controlled by the injection rate or the injection pressure. Fig. 1 illustrates the four different injection schemes applied in this study. Fig. 1(a) shows a constant high injection rate over time whereas Fig. 1(b) shows an alternating high–low injection rate using the same high injection rate of 100 mm^3/s and zero for the low injection rate. Fig. 1(c) shows the injection pressure–time curves under a constant pressurization rate until breakdown where MBP can be obtained. In Fig. 1(d), the maximum injection pressure in each loading-unloading cycle was below the MBP and specimen may fail after a certain number of cycles or may not fail depending on the magnitude of the maximum injection pressure. The two injection schemes in (a) and (c) correspond to conventional HF with MBP and the other two in (b) and (d) correspond to CHF with CBP.

3. Materials and methods

3.1. Experiment setup

A triaxial hydraulic fracturing test cell with an injection pressure capacity of 35 MPa, a maximum confining pressure of 20 MPa, and a maximum sample diameter of 50 mm was employed for testing cylindrical core samples. Fig. 2 shows the experiment setup. Nozzle plates with nipples are installed at the top and bottom of the specimen. Fluid is pumped by controlling the booster and is injected from the bottom of the testing sample. Embedded O-rings installed in the nozzle plate engage under applied vertical pressure and prevent fluid leakage. Zhuang et al.[40] describes the equipment in more detail. The hydraulic pump injects fluid under a controlled pressurization rate, injection rate, or a combination of the two. The pressurization can be applied in various ways, including linear and trigonometric functions, among others.

3.2. Testing material

Some granites contain three orthogonal cleavage planes: the rift (R), grain (G), and hardway (H) planes. The planes fail most easily under tension and they are identified by mineral properties, tectonic foliation and cracks.[41] The anisotropy of granite correlated to its microscopic fabric has been well addressed.[42] Pocheon granite obtained from a quarry in the Pocheon area in South Korea was used for the experiments. Polarizing microscopy images indicate that this rock has a dominant grain size of 0.3–3.0 mm. The cylindrical specimens tested were cored perpendicular to each of the R, G, and H planes, and were thus labeled R, G, and H, respectively, according to the cleavage plane lying on the circular plane as shown in Fig. 3. The rift plane has the highest microcrack density and therefore the lowest tensile strength; it also has the highest hydraulic conductivity. The measured average Brazilian tensile strengths (BTS) for these planes are 6.1, 8.2, and 8.8 MPa. For details of the physical and mechanical properties of Pocheon granite we refer to Zhuang et al.[40,43] All cylindrical specimens are 50 mm in diameter and 100 mm long, with an 8 mm-diameter hole through the center axis. The top and bottom surfaces of the specimens were ground.

3.3. Testing procedure

We compared continuous and cyclic injections in two sets of hydraulic fracturing tests. One set employed a confining pressure of 5 MPa and water was injected at a pressurization rate of 0.1 MPa/s, while the other set used a confining pressure of zero and water was injected at an injection rate of 100 mm^3/s. The vertical stress was fixed to 25 MPa in all tests to prevent leakage from the top and bottom surfaces of the specimen connected to the nozzle plates. First, the average MBP for each type of specimen during continuous injection was measured. Then the cyclic injection tests were employed to a maximum injection pressure equal to approximately 70%–101% of the average MBP.

After HF testing, each specimen was removed from the cell for X-ray CT scanning to observe the induced internal fractures. An open X-ray source tube with a 5 μm focal spot was used to acquire high-resolution images in combination with a flat panel detector having 2048 × 2048

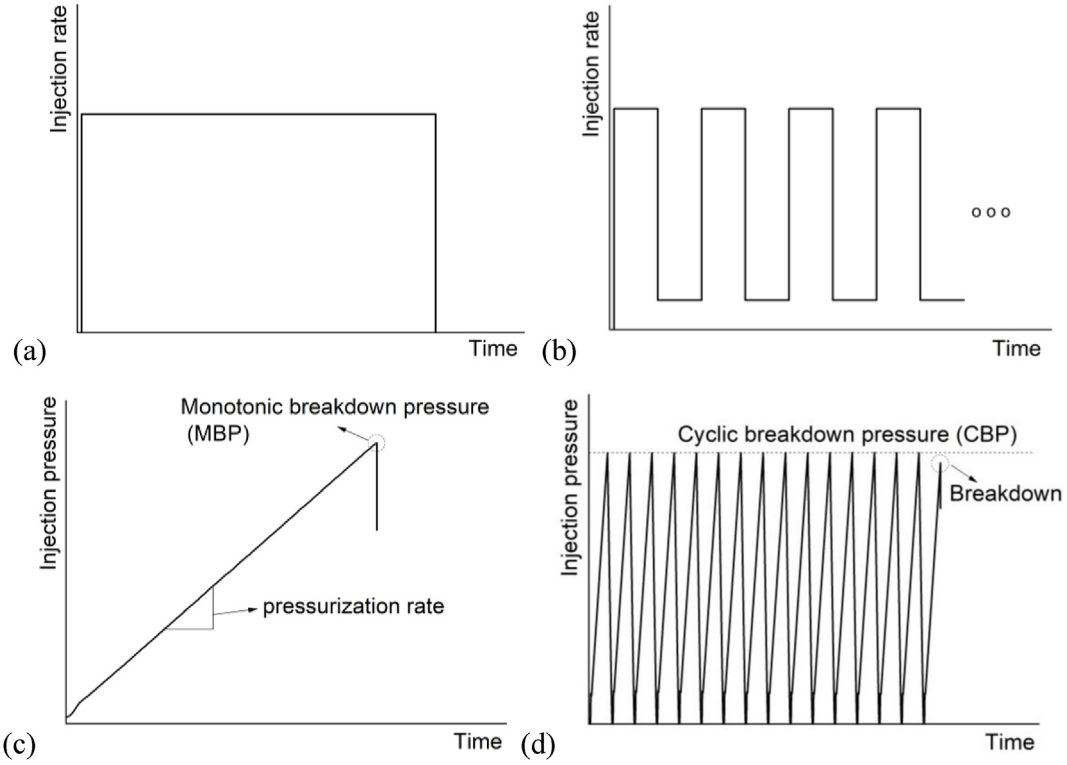


Fig. 1. Four different injection schemes: (a) and (b) injection rate control with constant and cyclic injection rate, respectively; (c) and (d) pressurization rate control with constant and cyclic pressurization rate, respectively. The pressurization rate is defined as the ratio of injection pressure increase measured from boreholes and the time increase. In (d), the maximum injection pressure is determined referring to the monotonic breakdown pressure (MBP) of granite sample fractured by the scheme in (c), and the pressure is termed as cyclic breakdown pressure (CBP) if breakdown occurs.

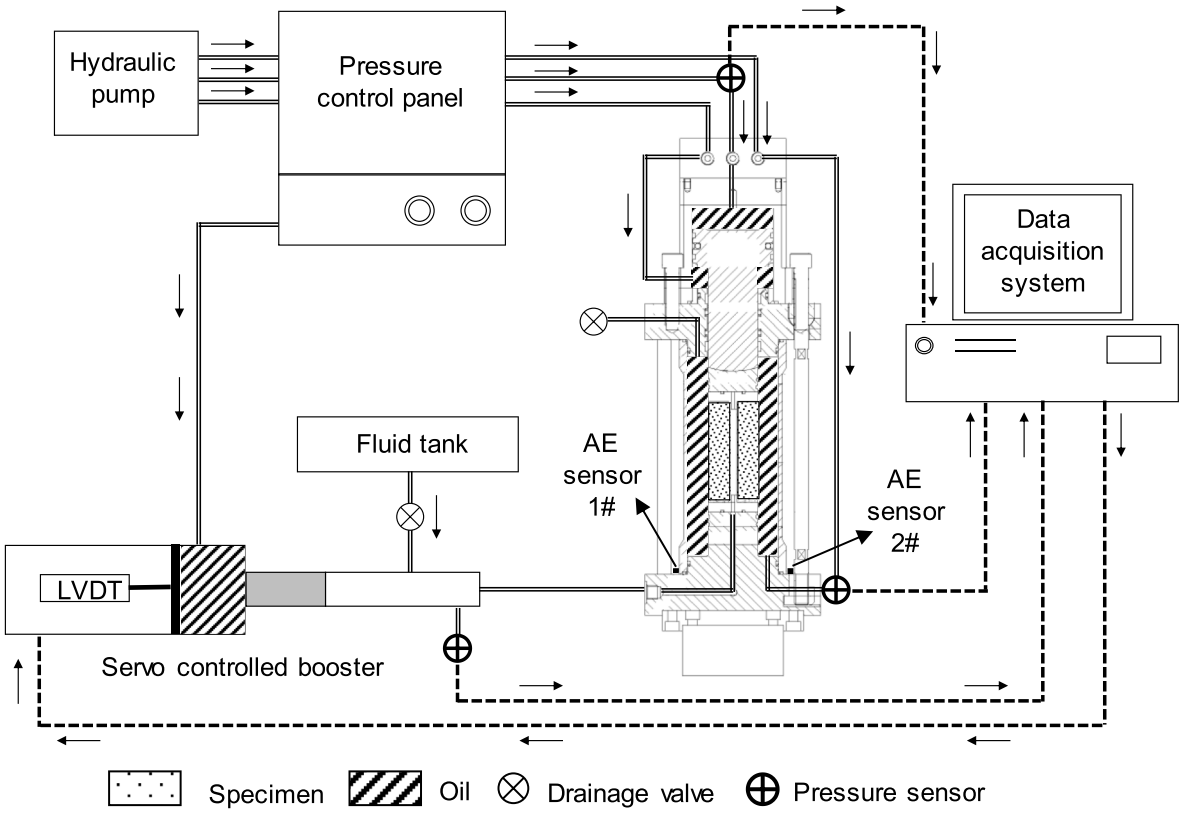
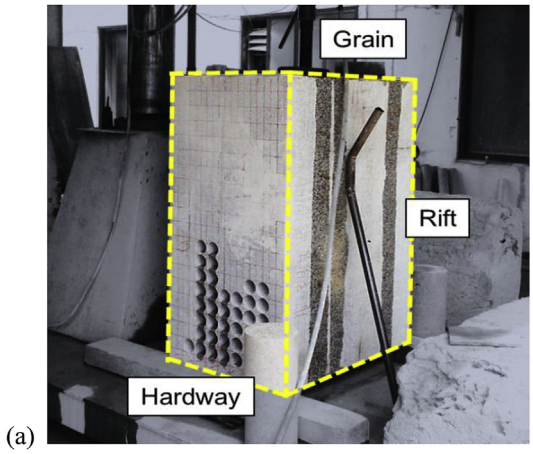
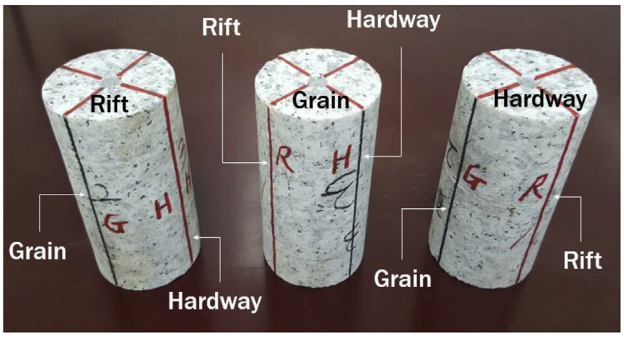


Fig. 2. Sketch of experiment system (modified from Zhuang et al., 2019).[38] Fluid is injected from the bottom of the specimen. We installed two AE sensors at the pedestal with fixed positions.



(a)



(b)

Fig. 3. Granite samples: a) coring from a granite block, b) from left to right: R, G and H samples with different relative positions of cleavage. R, G and H were named by the cross section plane of the sample.

pixels. The resulting CT images of the granite specimen have a pixel pitch of 0.058 mm. A 3D median filter (3 × 3 × 3 kernels) was applied to all acquired images prior to fracture segmentation. Image processing before the fracture segmentation was applied using embedded functions in VGStudio Max software (ver. 3.1) provided by the Volume Graphics GmbH.

3.4. Acoustic emission monitoring

AE could be monitored only in the injection rate-controlled tests. Two AE sensors were installed on the pedestal of the testing equipment during some of the tests to evaluate the seismicity induced by fracturing. The Nano 30 AE sensors have a resonant response of 300 kHz and an operating frequency range from 125 to 750 kHz. We acquired AE data using the AEwin software provided by the MISTRAS group.[44] After testing, the background noise was determined using the pencil-lead break test and the threshold amplitude was set to 40 dB for all tests. AE signals were amplified using a 40 dB preamplifier. Although energy attenuation likely lowered the measured amplitudes of induced AE in comparison with those that would have been measured directly on the specimen, a valid comparison of amplitudes of AE for cyclic and continuous injection is possible, as both were assessed under the same conditions.

4. Experimental results and analysis

4.1. Pressurization-rate-controlled HF tests

4.1.1. Breakdown pressure

Hydraulic fracturing tests with continuous injection were conducted on four R specimens, twenty G specimens, and five H specimens to obtain their average MBPs. Fig. 4 compares the individual and average

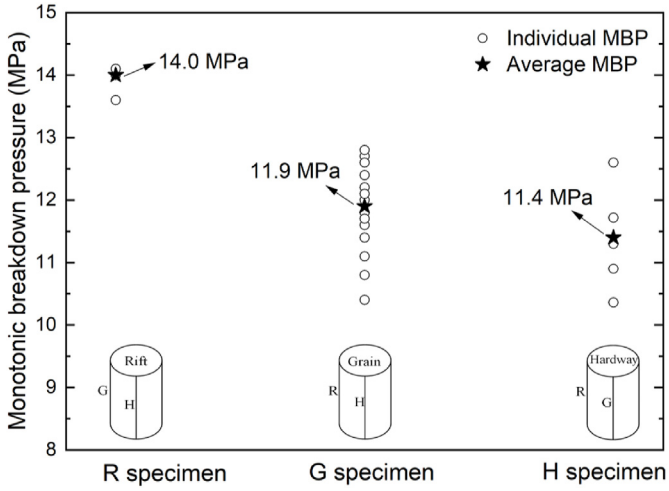


Fig. 4. Individual and average MBP for R specimens (4 cases), G specimens (20 cases), and H specimens (5 cases). The x-axis has no physical meaning but shows the three different types of specimens. Some data points overlapped due to their very close values.

BPs for the three specimen types. The average MBP, denoted by P_b^{avg}, is 14.0, 11.9, and 11.4 MPa, with a standard deviation of 0.24 MPa, 0.71 MPa and 0.85 MPa for the R, G, and H specimens, respectively. The intrinsic heterogeneity of the specimens inevitably caused their MBPs to vary even under identical test conditions. Despite this, the average MBP for each of the three types differed significantly due to the different tensile strengths of the three cleavage planes. The G and H specimens predominantly failed along the rift plane, as the rift plane has the lowest BTS among the three planes. The R specimens failed along the grain plane, which has lower BTS than the hardway plane. After testing, the locations of hydraulic fractures were confirmed by CT images.

The average MBP for the twenty G specimens and five H specimens combined was 11.8 MPa. This combination of results is valid given that the fractures were generated consistently on the rift plane. We performed CHF tests on nine G and nine H specimens, which must be selected based on their cleavage planes in order to minimize errors originating from the anisotropy of the granite. Fig.

5 shows the maximum injection pressure P_{max}, as well as the ratio P_{max}/P_b^{avg} versus the number of cycles for cyclic injection. The results show that 12 cases failed between 2 and 65 cycles, with the maximum injection pressures ranging from 79% to 101% of P_b^{avg}.

Six cases (three G specimens and three H specimens) did not suffer breakdown, as shown in Fig. 5. For example, one G specimen did not fail at 105 cycles when the maximum injection pressure was reduced to 9.0 MPa, which is approximately 76.3% of P_b^{avg}. Similarly, an H specimen withstood 150 cycles at the maximum injection pressure of 8.86 MPa, which is approximately 75.1% of P_b^{avg}. Note that some cases could not be tested using a larger number of cycles due to the limited capacity of the booster for pumping water. Later, the pumping system and testing program were improved to avoid this problem. In the second set of experiments under injection rate control, we had all of the samples tested until fracturing. In spite of the above limitation, comparison of the cases with and without breakdown indicates that more injection cycles are needed to break down the specimen when the maximum injection pressure is lower. If the pressure is too low relative to the monotonic BP of the specimen (e.g., lower than 80%), cyclic injection may not lead to macroscopic failure.

The H specimen that remained intact after 150 cycles was retested under continuous injection until it broke down. Its MBP was 10.9 MPa, approximately 7.6% lower than the average MBP of 11.8 MPa. This suggests that the specimen could have sustained some localized internal damage during cycling despite total failure not being achieved. We analyzed this hypothesis further based on the CT observations of the

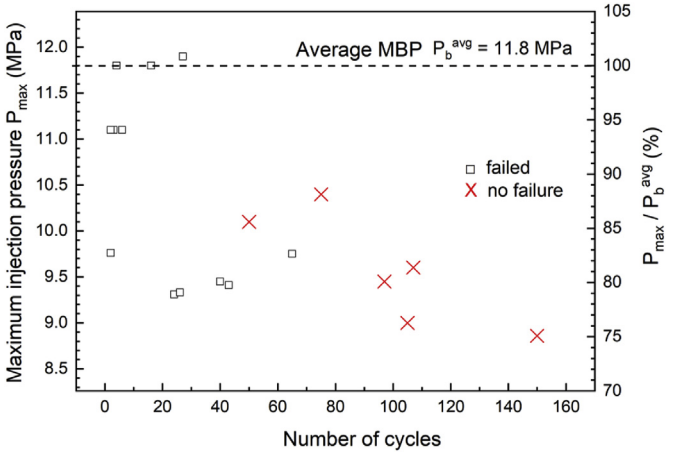


Fig. 5. Maximum injection pressure and corresponding P_{max}/P_b^{avg} plotted with respect to number of injection cycles. P_{max} is the maximum injection pressure in the borehole, P_b^{avg} is the MBP averaged on 20 G specimens and 5 H specimens.

hydraulic fractures in the next section.

4.1.2. CT observations of hydraulic fractures

Typical two-dimensional CT images and diagrams of the fractures are shown in Fig. 6. Some very fine fractures that are smaller than the image resolution of 0.058 mm per pixel may have been omitted. The line thicknesses in the diagrams represent the sizes of the fracture apertures. Since fractures exist in three dimensions, their properties and locations vary in different cross-sections through the specimen.

Fig. 6(a and b) show images of H and G specimens after continuous injection with MBPs of 12.6 and 12.1 MPa, respectively. Branching wing fractures were generated subparallel to the rift plane. Fig. 6(c and d) show images of two H specimens that failed after 43 and 65 cycles of injection, respectively. The fractures appear more complex and show a wider damage than those generated during continuous injection. Because they are branched, the fractures tend to have a wider spread than the fractures induced by continuous injection, although some are short and do not extend to the sample surface. The fractures presented in (c) and (d) are thinner than the biwing fractures in (a) and (b).

Fig. 6(e) shows an unusually durable H specimen that did not break down after 150 cycles of injection at a maximum injection pressure of 8.86 MPa and was subsequently fractured under continuous injection at a MBP of 10.9 MPa. The figure shows multiple fractures, especially in the lower half of the sample section. As explained above, although no breakdown was observed from the injection pressure curve, damage (e.g., small cracks) may have already accumulated within the specimen during cyclic injection. This may also account for the subsequent MBP being smaller than the average for the specimens that did not experience cyclic injection and for the greater complexity of the fractures than those induced by continuous injection.

Once injection starts, the injection pressure increases and water infiltrates into the rock matrix, thus opening and further developing existing microcracks in the granite specimen. Our past study on the same cylindrical granite specimens showed that at a constant injection rate of 100 mm^3/s, the initially dry specimens were not fully saturated until the breakdown, while for a long injection at relatively low injection rates, the specimens were finally completely saturated.[40] Moreover, numerical simulation of the same tests has shown that the infiltration depth from the borehole increases with increasing cycles of injection.[45] Therefore, we infer that cyclic injection allows injected water to infiltrate further into the specimen compared to continuous injection, where specimens were fractured relatively quickly (with breakdown within ~120 s). Also, slow injection allows chemical reaction of silica compounds to take place and lower the fracture strength. Hydraulic fractures are thus likely to form at multiple weak locations

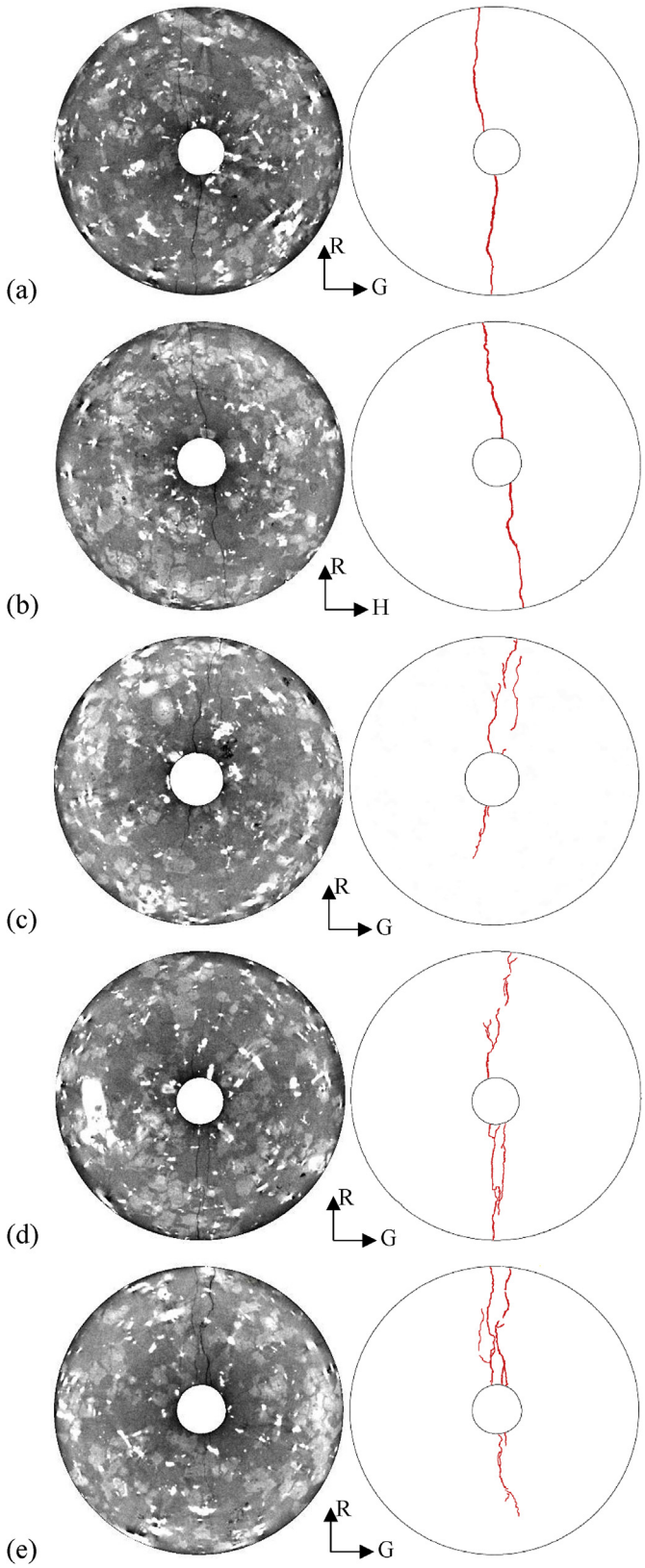


Fig. 6. CT images of granite specimen cross-sections (left) with accompanying sketches of hydraulic fractures (right) for different testing cases: (a) H specimen fractured by continuous injection, (b) G specimen fractured by continuous injection, (c) H specimen fractured by cyclic injection at 43 cycles, (d) H specimen fractured by cyclic injection at 65 cycles, and (e) H specimen experiencing 150 cycles of cyclic injection and further fracturing by continuous injection.

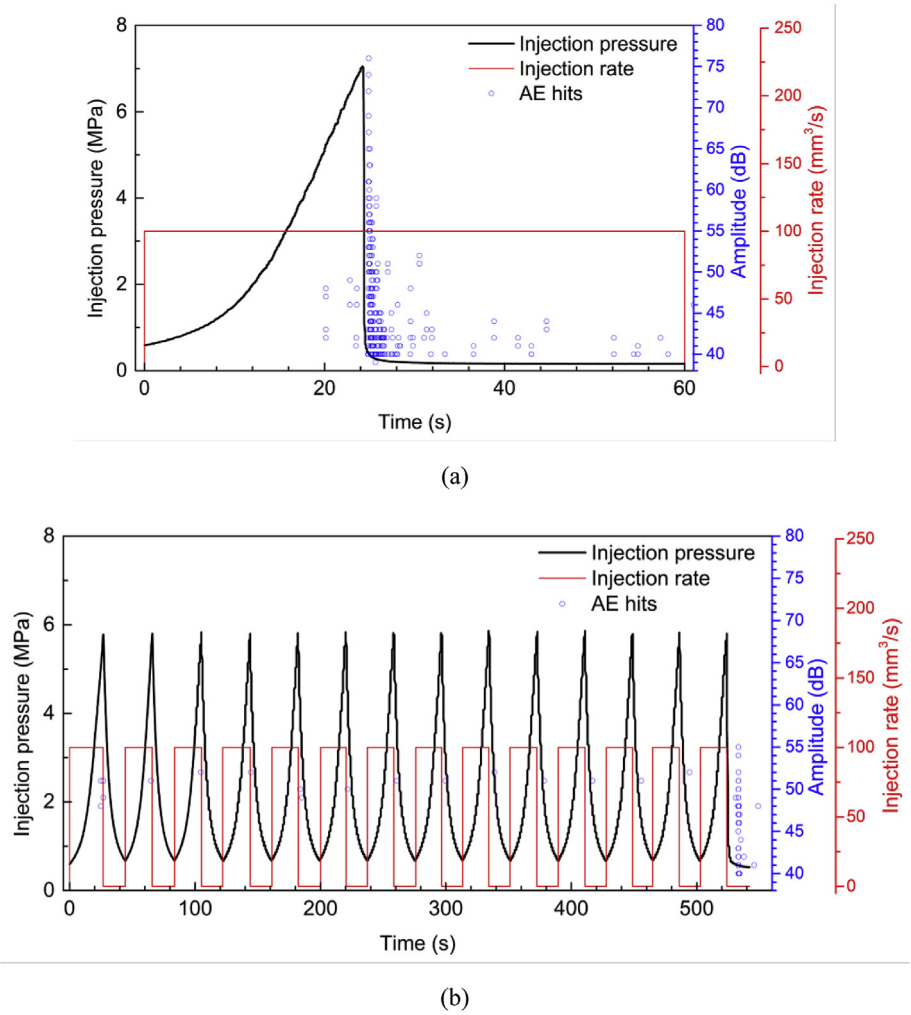


Fig. 7. Injection pressure-time curve, injection rate-time curve, and amplitude of induced AE hits for (a) continuous injection (HF-1#), and (b) cyclic injection (CHF-1#). One circle represents one AE hit.

inside the granite specimen, e.g., at long microcracks and the weak bonds between grains. As a result, cyclic injection is more likely to generate multiple fractures than the single fractures induced by continuous injection.

4.2. Injection rate-controlled HF tests

4.2.1. Breakdown pressure and AE amplitude

Fracturing under injection rate control was conducted on the H specimens using both continuous and cyclic injection. First, the average MBP, P_b^{avg}, at a constant injection rate of 100 mm^3/s and zero confining pressure was measured for 12 specimens to be 6.90 MPa with a standard deviation of 0.48 MPa. We then conducted subsequent cyclic injection tests to examine the maximum injection pressures, P_{max}, which yielded values that were 75%–95% of P_b^{avg}. Depending on the maximum injection pressure, each loading–unloading cycle lasted 15–45 s.

Fig. 7(a and b) show typical curves of the injection pressure evolution and induced AE hits with injection time and their amplitudes for continuous and cyclic injection, respectively. The maximum amplitude measured for three continuous injection tests varied from 61 to 76 dB, with an average of 67.0 dB, while it varied from 49 to 55 dB with an average of 53.3 dB for seven cyclic injection tests. Therefore, cyclic injection on average reduced the maximum amplitude of induced AE hits by 13.7 dB.

This reduction in AE could be attributed to cyclic injection using a lower maximum pressure compared with the MBP. Fig. 8 shows the

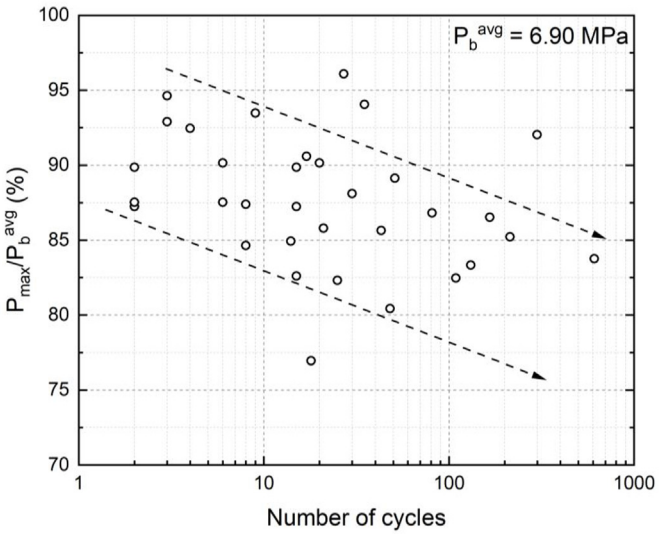


Fig. 8. Number of cycles required for breakdown under different P_{max}/P_b^{avg} ratios for a total of 34 cases of cyclic injection test. P_{max} is the maximum injection pressure in the borehole, P_b^{avg} is the average MBP measured on 12 specimens during continuous injection. X-axis is shown in logarithmic scale. The dash lines show the trend of data that the number of cycles increases with decreasing P_{max}/P_b^{avg} ratio.

P_{max}/P_b^{avg} ratios for 34 cases of cyclic injection with respect to the number of cycles until breakdown. The cyclic P_{max} was generally 80%–95% of P_b^{avg} for 33 cases, and was 76% in the one remaining case. The number of cycles required for breakdown varied from 2 to ~600, but was mostly below 100. The varying properties of the specimens led to great variation in the number of cycles needed, even for similar injection pressures. Nevertheless, the number of cycles required for fracturing the 34 samples generally increased logarithmically with decreasing maximum injection pressure P_{max}. The trend of data is plotted in Fig. 8.

Two specimens did not break down after many cycles. One of them withstood over 1000 cycles with P_{max} = 4.73 MPa (68.7% of P_b^{avg}) and the other endured more than 400 cycles with P_{max} = 5.44 MPa (79.0% of P_b^{avg}). Similar to the pressurization rate-controlled HF tests, too low of a P_{max} relative to the monotonic BP implied that it was difficult to induce breakdown, as little damage accumulated in the specimen. Overall, the results show that cyclic injection can induce hydraulic breakdown with a maximum injection pressure above 80% of the estimated MBP. Therefore, breakdown pressure can be reduced by ~20% by using CHF rather than continuous injection for the given testing conditions in the current study.

4.2.2. Injectivity measurement

The injectivity index or productivity index is used to evaluate stimulation or production efficiency in the field.[46–49] We assume that any permeability enhancement will be reflected indirectly in the injectivity, so that higher rock permeability leads to higher injectivity. Note that the injectivity measured here is different from that used in field application. We define the injectivity as the injection rate divided by the plateau value of the injection pressure. The average injectivity value of intact specimens measured at low injection rates of 1, 5 and 10 mm^3/s is approximately 3.8E-3 $cm^3/s/MPa$.[40] After the hydraulic fracturing test, water was injected continuously into the fractured specimen at the same injection rate of 100 mm^3/s and under the same stress conditions as applied during fracturing.

The plateau pressure varied from 0 to 0.16 MPa for 10 specimens fractured by continuous injection, while it varied from 0.02 to 6.17 MPa for 25 specimens fractured by cyclic injection. We plotted injectivity of the fractured specimen against breakdown pressure (MBP and CBP) for all cases in Fig. 9. Completely fractured specimens exhibited injection

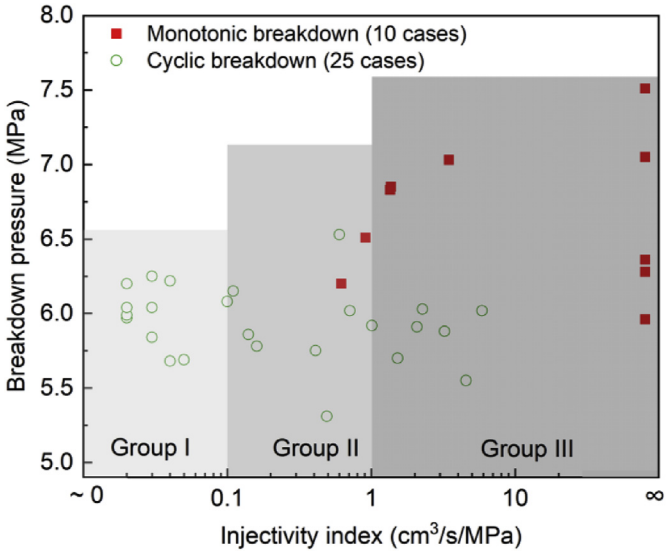


Fig. 9. Injectivity of specimens fractured by monotonic (10 cases) and cyclic (25 cases) breakdown plotted with breakdown pressures (MBP and CBP) of the corresponding specimens. Three groups (I, II and III) were divided indicating low, middle and high injectivities.

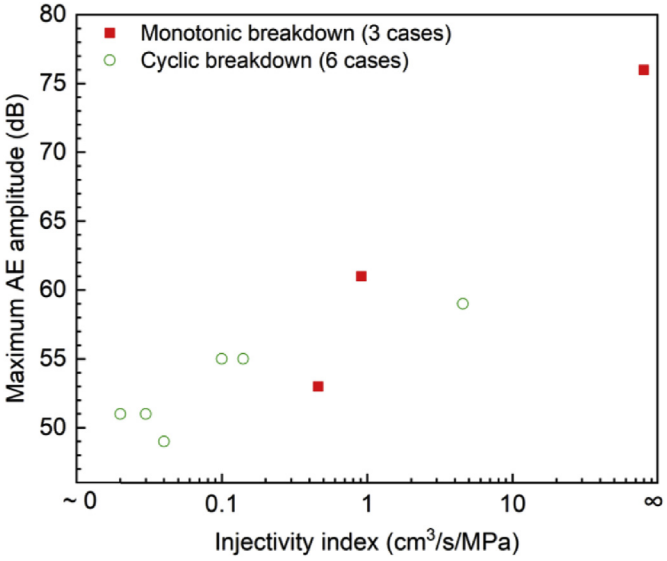


Fig. 10. Injectivity of specimens fractured by monotonic (3 cases) and cyclic (6 cases) breakdown plotted against the maximum AE amplitude observed during fracturing.

pressures of nearly zero, therefore the injectivity was calculated to be infinity (∞). We divide the results from tested samples into three groups based on injectivity magnitudes: group I (< 0.1 $cm^3/s/MPa$, low), group II (0.1–1.0 $cm^3/s/MPa$, middle), and group III (> 1.0 $cm^3/s/MPa$, high). The threshold injectivity was determined to be 0.1 $cm^3/s/MPa$ from low to middle and 1.0 $cm^3/s/MPa$ from middle to high, corresponding to the plateau pressure of 1.0 MPa and 0.1 MPa during the injection test. Injectivities of most samples fractured by conventional HF tend to be higher than those of the samples fractured by CHF. Among the 10 specimens that experienced monotonic breakdown, two fell into group II and eight into group III. However, the 25 CHF samples were distributed across all three groups: 10 in group I, 8 in group II, and 7 in group III. Over 60% of the fractured CHF samples had lower injectivity than the lowest injectivity of the HF samples. The lower injectivities obtained by CHF than those by monotonic HF is likely due to the lack of a single predominant fracture, as was generated through monotonic HF, which contributed greatly to the increase of fluid flow.

The injectivity index of six specimens fractured by CHF and three specimens fractured by HF are plotted against the maximum AE amplitude recorded during the fracturing in Fig. 10. We observe a trend that larger injectivities correspond to higher AE amplitudes, particularly for the monotonic breakdown cases. For a fractured rock specimen with one main tensile fracture, it is reasonable to infer that a higher injectivity or permeability relies on a larger aperture and a longer fracture, which accordingly result in higher strain energy release during tensile fracturing, corresponding to a larger magnitude seismic event.

5. Discussion

The ultimate goal of HF is to enhance hydraulic performance of the target reservoir. In this study, we considered two parameters to evaluate hydraulic performance of fractured specimens: the aperture of the hydraulic fractures and injectivity. The fractured specimens were scanned and the apertures of their hydraulic fractures were estimated based on CT images. Because the apertures of the induced fractures varied across the surface of each specimen, we selected their maximum values for comparison. For the injection rate-controlled tests, the maximum aperture for 10 cases of continuous injection ranged from 0.29 to 0.35 mm with an average of 0.33 mm, while for 25 cyclic cases the range was 0.17–0.35 mm with an average of 0.25 mm. However, there were many branching fractures and some fractures were too thin

to be observed because of the limited resolution of the CT images. Moreover, some fractures connected to each other in 3-D space, whereas others did not. Therefore, it is difficult to evaluate hydraulic performance enhancement of hydraulic fracturing by using only the measured aperture in one section or plane. Injectivity is a more appropriate parameter than the mechanical aperture of fractures for evaluating hydraulic performance of fractured samples.

Based on the injectivity measurements of the fractured granite specimens, the hydraulic performance of cyclic injection failed to exceed the performance of continuous injection. In some cases, hydraulic fractures propagated to the surface of the specimen, while in others they were relatively short with small apertures. The monotonic fractured specimens showed high injectivity approaching infinity while specimens tested under CHF had low injectivity of 0.02 $cm^3/s/MPa$.[22] However, this result is limited to the specific cyclic injection scheme alternating between a single high injection rate and zero injection rate. Hydraulic fracturing experiments in the Äspö HRL have shown that a combination of cyclic hydraulic fracturing and pulse pressurization, which is defined as fatigue hydraulic fracturing (FHF), shows a better permeability enhancement effect than those of both CHF and conventional HF.[11] CHF allows rock fatigue through frequent pressurization and depressurization phases. FHF is an enhanced process which involves progressive multiple flow rate cycles and high-pressure oscillations at the peak value at each cyclic pressurization phase. Recent experimental work following this study has also shown that pulse pressurization helps to achieve substantially improved permeability enhancement results on the same Pocheon granite.[24,27]

This study demonstrates the reduction of breakdown pressure and induced seismicity through applying CHF on intact granite core specimens. The applied stress conditions indicated that tensile fracturing is dominant, which was confirmed by additional AE monitoring analysis of crack types.[26] Tensile fracturing occurs at injection points or well sections where fluid is pumped at high pressure into the rock system. When the hydraulic energy is later distributed in the diffusion process, shear fracturing can also develop. In addition, aseismic opening and slip can play a role in the overall energy conversion process from hydraulic to fracture surface energy.

We use the AE maximum amplitude to evaluate the magnitude of induced seismic events during hydraulic fracturing to test if CHF mitigates induced seismicity. AE amplitude can indicate relative energy values because the energy is proportional to the squared amplitude. Calibration of AE sensors is necessary for finding absolute energy values. For example, ball drop experiment results have shown that the amplitude from the broadband AE sensors of 1 mVs is calibrated to account for an absolute energy value of 33 μJ in the case of red Aue granite from Erzgebirge Germany.[50] In combination with the pencil lead break test and ball drop test, AE monitoring allows one to estimate absolute values of radiated seismic energy. Similarly, the AEwin software used for data acquisition also provides the absolute energy of an AE event (MISTRAS Group Inc., 2014).[42] This value is obtained by integrating the square voltage divided by the reference resistance over the time duration of the waveform, with units of atto-Joule. In this study, calibration of AE amplitude was not fulfilled and the total number of AE sensors were insufficient to identify possible AE event locations. Improvement of AE monitoring and further analysis of the hydraulic energy and seismic energy during CHF on the same granite are currently ongoing.[25–27]

6. Conclusions

CHF is a promising alternative to conventional HF that can mitigate induced seismicity. In contrast to continuous injection, for which the injection rate increases monotonically or remains constant, cyclic injection alternates between high and low injection rates to reduce the fluid pressure, particularly at the fracture tip. We demonstrate this method here through laboratory hydraulic fracturing tests on intact granite cores. In the tests, water injection was controlled by injection rate or pressurization rate.

(1) Experiment results show that CHF generally reduces the BP by ~20% compared with the MBP of continuous injection. Accordingly, the maximum amplitude of acoustic emission during CHF is reduced by an average of 13.7 dB.
(2) CT observations show that CHF tends to produce complex and branched fractures, whereas conventional HF with continuous injection usually produces a single main fracture with little branching, particularly in specimens fractured by injection under a constant pressurization rate.
(3) The average values of maximum aperture of hydraulic fractures and injectivity are smaller for CHF samples than those for conventional HF samples. The propagation length, fracture aperture and measured injectivity varies greatly among the tests with CHF.

The conclusions above are limited to hydraulic fracturing of intact granite samples. Future work is necessary to verify the effect of cyclic injection on shear-dominated failure of rock mass with pre-existing discontinuties tested under true triaxial stress conditions. Specimens containing natural pre-existing fractures or artificial discontinuities can reasonably be used to simulate in situ hydroshearing.

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